

0. BIG PICTURE — DFD / DPP / PPP

Data Follow Distributions → Distributions Provide Probabilities → Probabilities Permit Prediction

1. KEY TERMS (EASY RECALL POINTS)

Term	Meaning
Probability	Likelihood / chance of a particular event
Statistics	Descriptive: summarize data. Inferential: generalize sample → population
Population	Well-defined collection of objects
Sample	Subset of the population
Experiment	Action/process whose outcome is uncertain
Sample space S	Set of all possible outcomes
RV X	Rule that assigns a number to each outcome in S
E[X] = μ _X	Mean of X (long-run average / center of mass of the pmf)
pmf p(x)	P(X = x) for a discrete RV
cdf F(x)	Cumulative Distribution Function: F(x) = P(X ≤ x)

Sample-exam recall: “cdf” = cumulative distribution function. Write the full phrase.

2. DESCRIPTIVE STATISTICS (ALWAYS SORT FIRST)

- $$\bar{x} = (\sum x_i) / n \quad | \quad S_{xx} = \sum (x_i - \bar{x})^2 = \sum x_i^2 - (\sum x_i)^2 / n$$
$$s^2 = S_{xx} / (n - 1) \quad | \quad s = \sqrt{s^2} \quad (\text{sample; use } n-1, \text{ not } n)$$
- **Median:** middle value after sorting. Even n → average the two middle values.
 - **Q1 / Q3:** medians of the lower / upper half. Even n: split into two equal halves, then take those medians. Review n=20 → Q1 = 6.
 - **α% trimmed mean:** drop α% of observations from *each* tail, then average the rest. 10% trim, n=10 → drop 1 from each end (10%×10=1).
 - **Histogram bins:** # bins ≈ √n. Interval notation: [a, b] inclusive; (a, b] means a < x ≤ b.
- If they give $\sum x$ and $\sum x^2$, do *not* re-sum the list. Use $S_{xx} = \sum x^2 - (\sum x)^2 / n$.

3. COMBINATORICS — ORDER VS NOT, REPEATS VS NOT

Situation	Formula	When
Permutation P(n,k)	$n! / (n-k)! = n(n-1)...(n-k+1)$	Order matters , no repeats (passwords with unique letters, lined-up crews)
Combination C(n,k)	$n! / [k!(n-k)!] = P(n,k) / k!$	Order does not matter (committees, choosing which rebars)
With replacement / repeats OK	n^k	k slots, n options each (4-letter words, letters may repeat)

5-letter words, no repeated letters, from 26: $P(26,5) = 26 \times 25 \times 24 \times 23 \times 22 = 7,893,600$
4-letter words, repeats allowed: $26^4 = 456,976$

Exam test: “same items in different order count as different” → permutation or n^k . “Unordered / just a set of objects” → combination.

4. PROBABILITY RULES

$0 \leq P(A) \leq 1, \quad P(S)=1, \quad P(A^c)=1-P(A)$
 $P(A \cup B) = P(A)+P(B)-P(A \cap B)$
 A, B mutually exclusive $\Rightarrow P(A \cap B)=0 \Rightarrow P(A \cup B)=P(A)+P(B)$

Conditional · Joint · Marginal
 $P(A | B) = P(A \cap B) / P(B) = [P(B|A) P(A)] / P(B)$
conditional = joint / marginal rearrange: $P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$

Law of Total Probability

$P(B) = \sum_i P(B | A_i) P(A_i)$
 A_i must be mutually exclusive AND collectively exhaustive (a partition of S)
Bayes' Theorem (write this from memory — it is a recall question)
 $P(A_i | B) = P(B | A_i) P(A_i) / \sum_j P(B | A_j) P(A_j)$
 $= P(A_i \cap B) / P(B)$ “flip” from cause → effect to effect → cause

Independence

A, B independent $\Leftrightarrow P(A|B)=P(A) \Leftrightarrow P(A \cap B)=P(A)P(B)$
Mutually independent $A_1...A_k$: $P(\text{all occur}) = \prod P(A_i)$

Do not assume independence. Prove it with $P(A \cap B)=P(A)P(B)$, or the problem must say “independent / random / with replacement / large population.”

5. DISCRETE RVS — PMF, CDF, MEAN, VARIANCE

pmf: $p(x)=P(X=x), \quad 0 \leq p(x) \leq 1, \quad \sum p(x) = 1$
CDF: $F(x)=P(X \leq x) = \sum_{i \leq x} p(i)$ (right-continuous step function)
 $P(X > a) = 1 - F(a) \quad | \quad P(X \geq k) = 1 - F(k-1) \quad | \quad P(a \leq X \leq b)=F(b)-F(a-1)$
 $E[X] = \mu = \sum x p(x) \quad E[h(X)] = \sum h(x) p(x)$
 $V(X) = \sigma^2 = E[(X-\mu)^2] = E[X^2] - \mu^2 \quad SD(X)=\sigma=\sqrt{V(X)}$
 $E[X^2] = \sum x^2 p(x)$ (not $(E[X])^2$)

When is $E[h(X)] = h(E[X])$? Only if h is **linear**: $h(x) = a + bx$.
Then $E[a+bX]=a+b\mu$ and $V(a+bX)=b^2 V(X)$, $SD=b | \sigma$.
If h is quadratic / $1/X$ / etc., you **must** compute $\sum h(x)p(x)$. Do not plug μ into h.

Worked nonlinear: want $E[2X^2+3]$ and $V[2X^2+3]$. First get $E[X^2]=V(X)+\mu^2$, then $E[2X^2+3]=2E[X^2]+3$. For the variance, $2X^2+3$ is *not* linear in X, so $V(2X^2+3)=4 V(X^2)$, and $V(X^2)=E[X^4]-(E[X^2])^2$. Compute $E[X^4]=\sum x^4 p(x)$ from the pmf. Shortcut $V(aY+b)=a^2V(Y)$ is OK after setting $Y=X^2$.

6. BINOMIAL VS HYPERGEOMETRIC

	Binomial X ~ Bin(n, p)	Hypergeometric X ~ Hyp(N, M, n)
Story	# successes in n independent Bernoulli trials; p constant	# successes in a sample of n drawn without replacement from finite N that has M successes
pmf	$P(X=x) = C(n,x) p^x (1-p)^{n-x}$	$P(X=x) = [C(M,x) C(N-M, n-x)] / C(N,n)$
Mean	$\mu = np$	$\mu = n (M/N) = np$ with $p=M/N$
Variance	$\sigma^2 = np(1-p)$	$\sigma^2 = np(1-p) [(N-n)/(N-1)]$ FPC
Use if	with replacement, or N huge so p barely changes, or trials independent	finite population, without replacement, n not tiny vs N

Support: binomial $x=0,1,...,n$. Hypergeometric $\max(0, n-(N-M)) \leq x \leq \min(n, M)$.

Multivariate hypergeometric (several types)
 $P(k_1,k_2,...,k_m) = [C(N_1,k_1) C(N_2,k_2) \cdots C(N_m,k_m)] / C(N, n)$
 $N = \sum N_i, \quad n = \sum k_i$. Classic: 5+5+5 rebars, grab 6, want 2 of each: $C(5,2)^3 / C(15,6) = 1000/5005 = 0.1998$

Flu example (15 students: 4 sen, 5 jun, 6 frosh; pick 3):
 $P(\text{exactly 2 frosh}) = C(6,2)C(9,1)/C(15,3)=135/455=0.2967$
 $P(\text{all 3 same class}) = [C(4,3)+C(5,3)+C(6,3)] / C(15,3)=34/455=0.0747$
 $P(\text{one of each}) = C(4,1)C(5,1)C(6,1)/C(15,3)=120/455=0.2637$

7. JOINT RVS — INDEPENDENCE CHECK

Joint pmf $p(s,t)=P(S=s, T=t)$. Marginals: $p_S(s)=\sum_t p(s,t)$, $p_T(t)=\sum_s p(s,t)$.
Independent RVs $\Leftrightarrow p(s,t) = p_S(s) p_T(t)$ for **every** pair (s,t).

One counterexample kills independence. Review Ex. 3: $P(S=0)=0.25$, $P(T=0)=0.5$, product=0.125, but $P(S=0, T=0)=0.25 \neq 0.125 \rightarrow$ **dependent**. Matching marginals is not enough. Zeros in the joint table while both marginals are positive also scream dependence.

8. BAYES TREES (DO THIS ON EVERY WORD PROBLEM)

1. Define events. First branches = partition A_i with $P(A_i)$.
2. Second branches = $P(B|A_i)$ and $P(B^c|A_i)$.
3. Joint on each leaf: $P(A_i \cap B) = P(A_i)P(B|A_i)$.
4. $P(B)$ =sum of B-leaf joints. Then $P(A_i|B)$ =joint_i / $P(B)$.

Austine energy: $P(A_1, A_2, A_3) = (0.40, 0.35, 0.25)$, $P(B|A) = (0.30, 0.60, 0.50)$
 Joints: 0.12, 0.21, 0.125 $\rightarrow P(B) = 0.455$
 $P(A_2 \cap B) = 0.21$ | $P(A_1|B) = 0.12/0.455 = 0.2637$ | $P(A_2|B) = 0.4615$ | $P(A_3|B) = 0.2747$

Upgrade/stable Bayes (sample exam): $P(U) = 0.20$, $P(S|U) = 0.90$, and $P(S)$ from the table (450/525). Then $P(U|S) = (0.90)(0.20)/P(S) = 0.21$. Do not invent $P(S)$ if the table already gives it.

9. FILLING A CONTINGENCY TABLE

1. Write every known count. Translate each sentence into an equation.
2. "One-third of all dams are rural" $\rightarrow$ Rural = (1/3) Total, so Urban = (2/3) Total $\rightarrow$ Total = (3/2) $\times$ Urban.
3. Row/column totals are gold. Check that every row and column adds.
4. Conditional from a table: $P(\text{unstable} | \text{rural}) = (\# \text{ rural and unstable}) / (\# \text{ rural})$.

Sample exam rural row: 100, 20, 50, 5 (old-stable, old-unstable, new-stable, new-unstable). $P(\text{unstable} | \text{rural}) = 25/175 = 0.1429$.

10. ESTIMATING BIN(N, P) FROM DATA

If yearly counts of "contaminated lakes" look like $X \sim \text{Bin}(n, p)$ with n, p unknown:

$$\bar{x} \approx np \quad s^2 \approx np(1-p)$$

$$\hat{p} = 1 - s^2 / \bar{x} \quad \hat{n} = \bar{x} / \hat{p}$$

Sample: $\bar{x} = 89.7$, $s = 9.044$, $s^2 = 81.79 \rightarrow \hat{p} = 1 - 81.79/89.7 = 0.08819 \rightarrow \hat{n} = 89.7/0.08819 \approx 1017$.

Extra-credit attack on binomial: lakes are not independent (spatial clusters of mercury); population of lakes is finite (hypergeometric); p may change year to year; n may not be constant; sampling design may not be simple random trials.

11. EXAM PLAYBOOK (MIRRORS THE SAMPLE MIDTERM)

Section	What they want
Recall	cdf name; full Bayes formula; $E[h(X)] = h(E[X])$ iff h linear
Manipulate a pmf	List $(x, p(x))$; compute $C = h(X)$ at each x ; $E[C] = \sum h(x)p(x)$; $V(C) = E[C^2] - (E[C])^2$; $SD = \sqrt{V}$. Keep units (\$M, etc.).
Counting	Usually multivariate hypergeometric: favorable C's on top, C(N,n) on bottom
Descriptive + n,p	mean, trimmed mean, median, s; then method of moments for binomial n,p
Joint table + Bayes	fill missing cells $\rightarrow$ conditional from table $\rightarrow$ Bin vs Hyp for "k of n dams" $\rightarrow$ Bayes with a new percentage

Binomial vs hypergeometric on "5 of 8 older dams stable": Exact: $N=250$ older, $M=200$ stable, $n=8$, $x=5 \rightarrow$ Hyp. Approx: $p=200/250=0.8 \rightarrow \text{Bin}(8, 0.8)$. Sample answers: 0.1468 (binom) or 0.1471 (hyper). If N is large vs n , say you used binomial as approximation and why.

12. CALCULATOR & PRESENTATION

- nCr : combinations. nPr : permutations. Don't mix them.
- Carry extra digits; round only the boxed answer (3–4 significant figures unless they want a percent).
- Percents vs decimals: $21\% = 0.21$. Stay consistent inside a calculation.
- Write the distribution name and parameters before plugging in numbers.
- If sampling without replacement from a listed finite set, it is hypergeometric / multivariate hypergeometric — not binomial — unless you explicitly approximate.

13. MEMORY DUMP — WRITE THESE FIRST ON SCRAP PAPER

$$P(A|B) = P(A \cap B) / P(B)$$

$$\text{Bayes: } P(A_i|B) = P(B|A_i)P(A_i) / \sum P(B|A_j)P(A_j)$$

$$\text{Independent} \Leftrightarrow P(A \cap B) = P(A)P(B) \Leftrightarrow p(s,t) = p(s)p(t) \quad \forall \text{ cells}$$

$$E[X] = \sum xp(x) \quad E[h(X)] = \sum h(x)p(x) \quad V = E[X^2] - \mu^2$$

$$E[a+bX] = a + b\mu \quad V(a+bX) = b^2\sigma^2 \quad \text{linear only}$$

$$\text{Bin: } C(n,x)p^x(1-p)^{n-x} \quad \mu = np \quad \sigma^2 = np(1-p)$$

$$\text{Hyp: } C(M,x)C(N-M, n-x) / C(N,n) \quad \mu = nM/N \quad \sigma^2 = np(1-p)(N-n)/(N-1)$$

$$\text{Multi-hyp: } \prod C(N_i, k_i) / C(N,n)$$

$$s^2 = [\sum x^2 - (\sum x)^2/n] / (n-1) \quad \hat{p} = 1 - s^2/\bar{x} \quad \hat{n} = \bar{x}/\hat{p}$$